

FrostCart — Cart UI/UX & Boundary Specification

Strict cart behavior, visual rules, motion, accessibility, and failure cases for the prototype.

Purpose

The cart must use familiar e-commerce behavior. The goal is immediate comprehension: product added → visible feedback → cart count updates → cart can be reviewed. Keep it simple enough for a 3-hour prototype.

1. Cart placement and trigger

- Desktop: cart icon in the top-right header.
- Mobile: cart icon remains in the top-right header.
- Badge shows quantity of items; hide badge at zero.
- Cart icon/button minimum 44 × 44 px.
- Clicking cart opens a right-side drawer on desktop and a bottom sheet/full-height panel on mobile.
- Do not replace the familiar cart icon with an unusual icon.

2. Add-to-cart interaction

Event	Required feedback
User taps Add	Button changes to ✓ Added briefly or becomes quantity control.
Cart count	Increments immediately; do not wait for animation.
Product card	Never jumps in height after adding.
Cart icon	Subtle 160–220 ms scale/badge animation.
Error	Show a short inline message; never silently fail.
Already in cart	Increase quantity or focus the existing cart line; do not create a duplicate line unnecessarily.

3. Cart drawer layout

Recommended order: Cart title → item count → product lines → subtotal → primary checkout CTA → secondary continue shopping action.

Element	Desktop	Mobile
Drawer width	400–440 px	Full width / bottom sheet
Title	20 px / 700	20 px / 700
Product title	15–16 px / 600	15 px / 600
Metadata	12–13 px	12 px
Price	16–18 px / 700	16–18 px / 700
Quantity controls	44 px target	44 px target

Element	Desktop	Mobile
Checkout CTA	Full drawer width	Full width, sticky near bottom

4. Cart item structure

[Product image] Brand / product name
 Pack size · optional variant
 ₹180
 [−] 1 [+] [Remove]

Keep quantity controls obvious. If quantity reaches zero, remove the line after a short state transition or ask only when the action is destructive and ambiguous. For a small grocery cart, direct removal is usually faster.

5. Cart state machine

State	UI
Empty	Centered cart icon + 'Your freezer is waiting.' + Browse frozen food CTA.
1+ items	Show item list and subtotal.
Updating	Disable only the affected quantity control briefly; preserve visible values.
Unavailable item	Mark unavailable; prevent checkout until resolved.
Maximum quantity	Show inline limit message; never silently cap without feedback.
Error	Inline retry/remove option; keep the rest of the cart intact.
Checkout unavailable	Explain why and offer continue shopping.

6. Strict boundary cases

- Double-click Add: do not accidentally add two items unless two distinct activations were intended; debounce the animation but preserve legitimate quantity changes.
- Rapid + taps: quantity should update predictably; animation must not queue dozens of effects.
- Rapid − taps: never go below 0.
- Remove while quantity is updating: resolve to the latest user action; do not resurrect a deleted item.
- Same product added twice: merge into one line with quantity 2 unless variants differ.
- Long product name: clamp to 2 lines.
- Missing image: use FrostCart placeholder.
- Very large price: prevent overflow; allow wrapping or compact formatting.
- Zero-price item in demo: display ₹0 honestly; do not invent a discount.
- Empty cart: checkout button must be disabled/hidden; show Browse CTA.
- Cart drawer open + viewport resize: recalculate width/height without horizontal scroll.
- 390 px: no clipped buttons; checkout CTA remains reachable.
- Keyboard: drawer must be focusable; Escape closes it; focus should return to the cart trigger.
- Screen reader: announce item count changes where practical; icon-only controls need accessible labels.
- Reduced motion: skip badge bounce and drawer slide; use fade/instant state.

7. Checkout boundary

- For this assignment, checkout can be a visual/demo action. Do not pretend a payment has occurred.
- If the button is non-functional, label it appropriately (e.g. 'Checkout demo') rather than claiming a real transaction.
- Do not display fake delivery times, live location, live inventory, or payment success unless actually implemented.

8. Cart motion language

- Drawer enter: 280–350 ms ease-out.
- Drawer exit: 200–250 ms ease-in.
- Badge count: 160–220 ms scale/opacity.
- Add confirmation: 160–220 ms.
- Quantity change: no large movement; update value instantly with subtle opacity/scale only.
- Do not animate every cart item simultaneously.

9. Cart copy

Non-empty: Your freezer picks

Empty: Your freezer is waiting.

Primary: Continue to checkout

Secondary: Keep browsing

Success: Added to your freezer

10. Cart QA checklist

- Add one item → count 1 → drawer contains one line.
- Add same item again → count 2 → one line quantity 2.
- Remove item → count decreases to 0 → empty state appears.
- Use +/– rapidly → no negative quantity, no duplicated lines.
- Open/close drawer repeatedly → no scroll lock bugs.
- Resize desktop ↔ mobile → no overflow.
- Refresh → if persistence is intentionally not implemented, cart resets; do not imply persistence.
- Keyboard-only navigation works.
- prefers-reduced-motion works.
- All controls have clear hover/focus/pressed/disabled states.